

SUMMARY

Pre-final year Engineering student with hands-on experience building production-grade full stack applications, REST APIs, and AI-powered systems. Proficient in Next.js, React, FastAPI, Python, and PostgreSQL. Passionate about shipping clean, scalable software — from responsive frontends and robust backends to LLM integrations and ML pipelines

WORK EXPERIENCE

- Data Engineering Intern | SPRINGER CAPITAL | [LINK](#)

January 2026 - April 2026

 - Built and orchestrated 2+ end-to-end ETL pipelines using Python, Apache Airflow, and PostgreSQL for structured data ingestion and processing
 - Designed data transformation workflows converting raw CSV data into analytics-ready Parquet and SQL datasets (Bronze → Silver architecture) & automated data validation and schema checks across 5+ pipeline stages, ensuring data quality and reliability
- Data Analytics Intern | INFOSYS SPRINGBOARD | [LINK](#)

October 2025 - December 2025

 - Analyzed 6.7M+ flight records using Python and SQL to identify delay patterns across weather, carrier operations, and seasonal trends
 - Built analytical queries and feature engineering pipelines to investigate data issues and generate insights, identifying 7.4× higher winter cancellations and 2.8× holiday delay impact

PROJECTS

- AutoFlow– Visual Workflow Automation Platform (Next.js, FastAPI, MongoDB) | [LINK](#)

August 2025 – February 2026

 - Built an AI-powered workflow automation platform enabling users to generate and execute workflows from natural language prompts using LLM-based planning, reducing manual setup effort by 70%
 - Developed a mobile responsive frontend with Next.js and React Flow, allowing users to visually design, edit, and execute automation pipelines with real-time feedback
 - Implemented scalable FastAPI backend with REST APIs, LLM integration, and strict validation pipelines to ensure reliable execution, debugging, and monitoring of automated workflows
- MedPrompt+– AI Health Assistant (Next.js, FastAPI, XGBoost) | [LINK](#)

December 2024 - February 2025

 - Built a full stack AI health assistant with 5 modular Next.js pages and a responsive dark/light dashboard, engineering a FastAPI backend with XGBoost/Scikit-learn risk prediction APIs serving Low/Moderate/High scoring in real-time
 - Designed an end-to-end RAG pipeline that ingests uploaded lab reports (PDF/image), extracts key vitals via OCR, generates structured medical summaries, and enables contextual LLM-powered Q&A over parsed documents
 - Integrated OpenAI across 4 LLM features — conversational health Q&A with memory, AI-generated 7-day food plans, lifestyle recommendations, and document summarization — all served through modular REST endpoints
- MotorGuard — IoT + AI Predictive Maintenance Platform (Next.js, FastAPI, Random Forest)

January 2026 - March 2026

 - Built an end-to-end IoT pipeline from ESP32 firmware (C) through a FastAPI backend to a Next.js dashboard — ingesting real-time motor telemetry over Wi-Fi JSON with 10+ REST endpoints for live monitoring, analytics, and model management
 - Engineered a Random Forest fault classifier across 5 fault classes with a 75% confidence gate and automatic rule-based fallback, computing health scores, efficiency metrics, and RUL estimates stored in MongoDB Atlas for trend analysis and model retraining
 - Integrated a Groq-powered RAG assistant over live motor records via endpoint, enabling contextual diagnostic Q&A on recent sensor history alongside a Next.js dashboard with real-time fault visualization and trend charts

SKILLS SUMMARY

- Frontend:

TypeScript, Next.js, React, React Flow, Tailwind CSS, HTML, CSS, JavaScript
- Backend:

FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB, Python, Docker
- AI / ML:

LLM Integration, RAG Pipelines, XGBoost, Random Forest, Prompt Engineering
- Data / Tools:

Git, GitHub, VS Code, JIRA, Vercel, Excel, Apache Airflow, ETL Pipelines, SQL, Pandas, NumPy

EDUCATION

- Christ College Engineering

B.E. Electronics and Communication Engineering ; GPA: 8.06

Machine Learning (MINOR) , English Proficiency: C1 (Advanced)

Achieved Top 2 rank in class (2 semesters)
- June 2023 - August 2027

Kerala, India

CERTIFICATES & ACHIEVEMENTS

- Generative AI Engineering (IBM) | [CERTIFICATE](#)

2× Hackathon Winner (AI/ML Based)
- December 2025